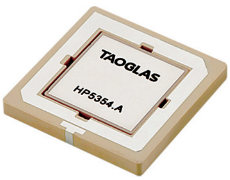
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Industry's first dual-band GNSS 'patch-in-a-patch' antenna

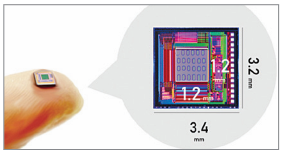
Taoglas has introduced the Inception HP5354.A, a GNSS L1/L5 low-profile antenna. The design embeds one antenna into another without physically stacking them, offering a dual-feed surface mount design (SMD) that reduces antenna height by 50%, decreases weight, and conserves space. This makes it suitable for GNSS applications that require precision and accuracy. It serves as a substitute for single-band surface mount patches, improving positioning accuracy from 3 metres to 1.5 metres without affecting dual-band L1/L5 performance. The antenna is used in applications including asset tracking, agriculture, industrial tracking, commercial drones, and autonomous vehicles. It maintains circular polarisation gain through a dual-feed design, ensuring performance stability even when the antenna requires de-tuning or in-situ adjustments.



Taoglas
https://www.taoglas.com

World's smallest high-resolution CMOS odour sensor

Aroma Bit, Inc. has introduced a prototype of the CMOS semiconductor e-nose type odour imaging sensor. The company claims to have developed a technology that accommodates 100 odour-receptor membranes within a 1.2mm x 1.2mm sensor area. The sensor's integration capability across various devices is expected to drive market expansion, especially in sectors requiring cost-efficiency and high volume. It is the smallest and densest sensor array for e-nose type odour sensors, facilitating integration into various devices. By depositing over 100 different types of odour-receptor membranes this sensor improves odour discrimination capabilities. This sensor lets electronics makers add features to gadgets that can check air quality or levels of freshness of food. Healthcare providers can use it to help diagnose diseases by analysing breath.



Aroma Bit, Inc.
https://www.aromabit.com

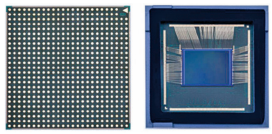
The innovative Greip dual-input audio module

USound has launched the Greip audio module in collaboration with OBO Pro 2 to improve TWS earphones and in-ear monitors (IEMs). It is a ready-to-use solution that includes an electrodynamic driver for bass and a MEMS speaker for clarity. The Greip module is intended for uses that require bass, active noise cancellation (ANC), and high-resolution audio. It is positioned to impact consumer electronics, automotive audio, and professional audio equipment significantly. Its design facilitates easy integration into various systems.

USound
https://usound.com

Image sensor for better smartphone photography

Samsung Electronics has introduced the mobile image sensor ISOCELL HP9. Its proprietary high-refractive microlens, made of a new material, enhances light-gathering capability by directing more light accurately to the corresponding RGB colour filter. This results in better colour reproduction and focus, with 12% better light sensitivity and 10% improved autofocus contrast performance compared to previous models. The HP9's optical format is suitable for telephoto modules, matching main cameras in image quality, autofocus, HDR, and fps. Powered by a remosaic algorithm, it offers 2x or 4x in-sensor zoom modes, achieving up to 12x zoom with a 3x zoom telephoto module, while maintaining image quality.



Samsung Electronics
https://www.samsung.com/in

SoC cuts EV design and manufacturing costs

Silicon Mobility, an Intel Company, has launched the OLEA U310 system-on-chip (SoC). The company claims it is the first solution combining hardware and software to meet the powertrain domain control needs in electrical architectures with distributed software. Its hybrid and heterogeneous architecture allows one unit to replace up to six standard microcontrollers, controlling an inverter, motor, gearbox, DC-DC converter, and on-board charger.